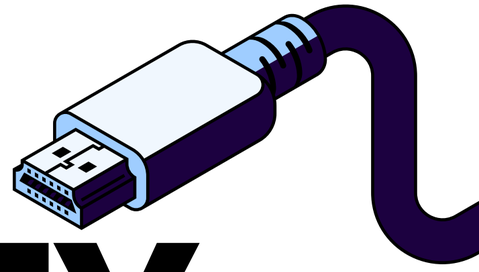
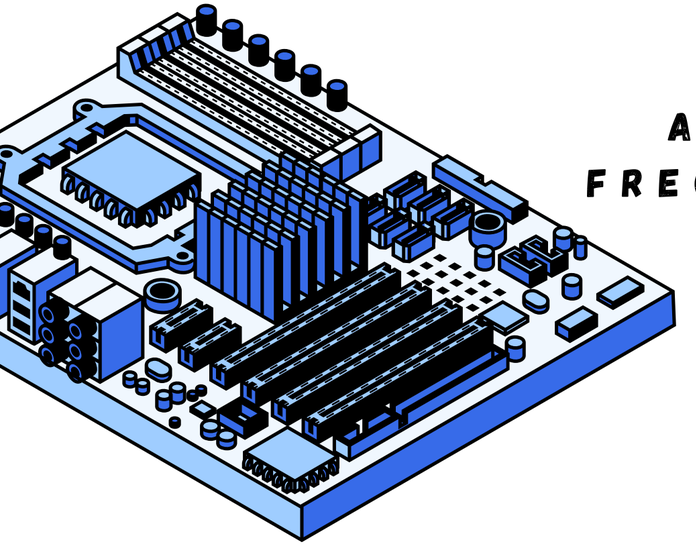
RFID BASED SHOPPING TROLLEY





AUTOMATED BILLING SYSTEMS USING RADIO FREQUENCY IDENTIFICATION (RFID) TECHNOLOGY

PROBLEM STATEMENT

LONG QUEUE

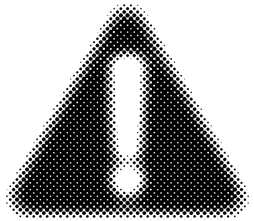
CUSTOMERS WAIT EXTENDED PERIODS AT CHECKOUT COUNTERS

SLOW PROCESS

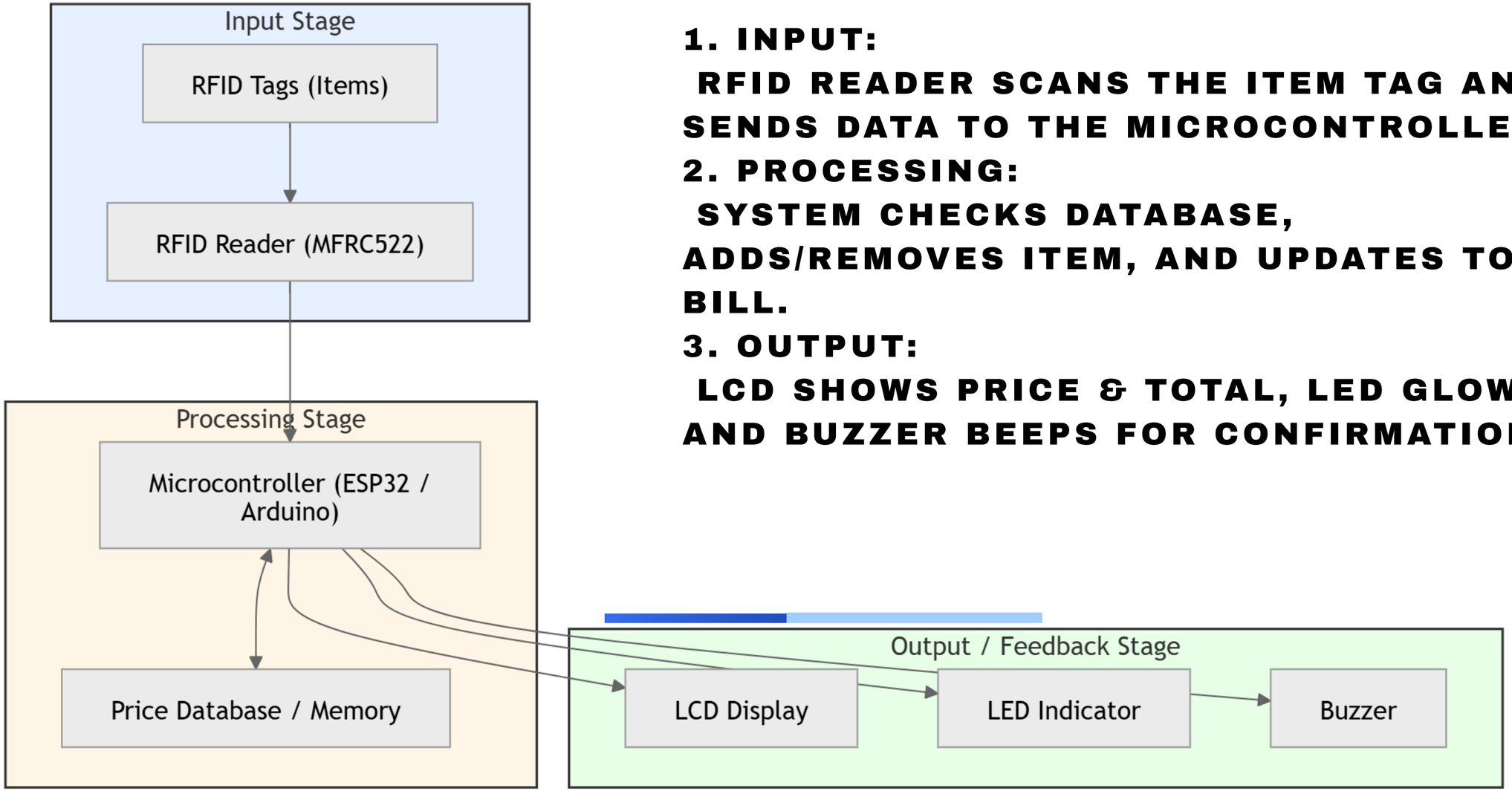
SLOW PROCESSING TRADITIONAL BARCODE SCANNING CREATES BOTTLENECKS DURING PEAK HOURS

MANNUAL SCANNING

MANUAL SCANNING LINE-OF-SIGHT REQUIRED, HUMAN ERRORS INCREASE WITH FATIGUE



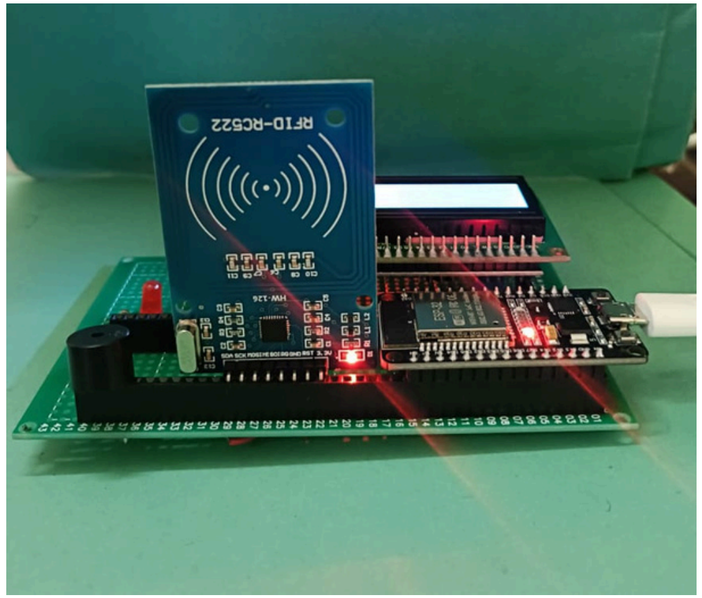
OUR SOLUTION



1. INPUT:
 RFID READER SCANS THE ITEM TAG AND SENDS DATA TO THE MICROCONTROLLER.
2. PROCESSING:
 SYSTEM CHECKS DATABASE, ADDS/REMOVES ITEM, AND UPDATES TOTAL BILL.
3. OUTPUT:
 LCD SHOWS PRICE & TOTAL, LED GLOWS AND BUZZER BEEPS FOR CONFIRMATION.

PROTOTYPE

Added:Laptop
Total:60000Rs
PY-8*12CM 2.54MM


RFID-RC522

Added:Top
Total:60500Rs
PY-8*12CM 2.54MM

HARDWARE COMPONENTS

ESP32

Main processor

RC522 Reader

RFID scanning

RFID Tags

Product IDs

LED & BUZZER

Indicator/Status feedback

16x2 LCD

Display output

Power Supply

System power

SOFTWARE COMPONENTS

EMBEDDED PROGRAMMING

ARDUINO IDE USED TO PROGRAM THE ESP32

MICROCONTROLLER LIBRARIES

MFRC522 LIBRARY FOR RFID READER COMMUNICATION

BACKEND DATA HANDLING

GOOGLE SHEETS API
WEB APP SCRIPT

FRONTEND (UI)

REACT.JS
NEXT.JS

Conclusion & Future Scope

CURRENT BENEFITS

→ SMART BILLING SOLUTRION
→ REDUCED CHECKOUT QUEUES

FUTURE ENHANCEMENTS

→ MOBILE APP INTEGRATION FOR SHOPPING LIST MANAGEMENT
→ ONLINE PAYMENT GATEWAY INTEGRATION
→ INVENTORY MANAGEMENT SYSTEM

Thank You